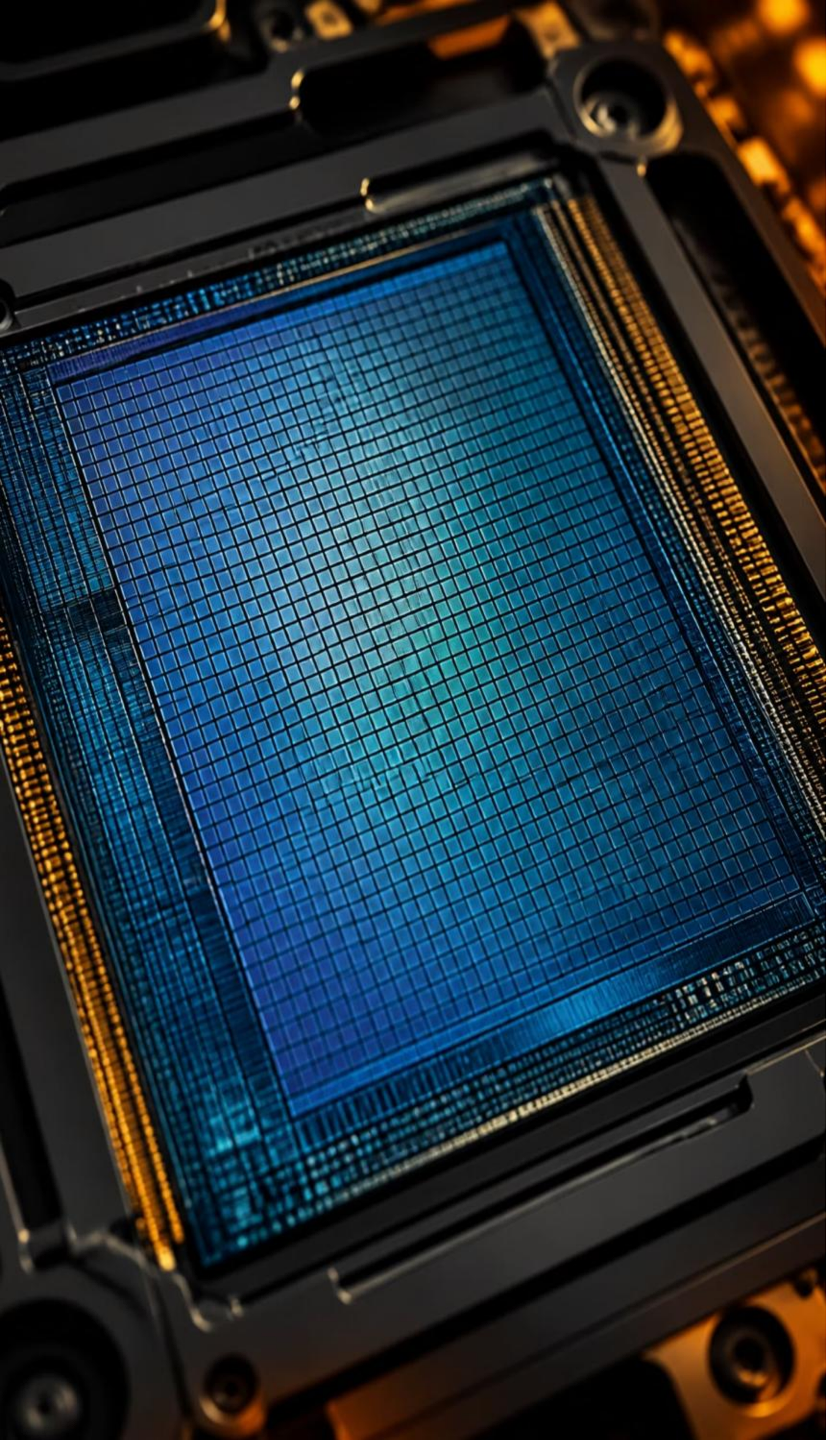


SIMATS ENGINEERING
Engineer to Excel
SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

The Anatomy of Clarity: Factors Affecting Image Quality

Guided by:
Dr. S. Yuvaraj

Presented by
M. Phanith Chandu
192425238

The Diagnostic Framework

Definition

Image quality = the exactness with which an object is represented in the image domain.

Success Criterion

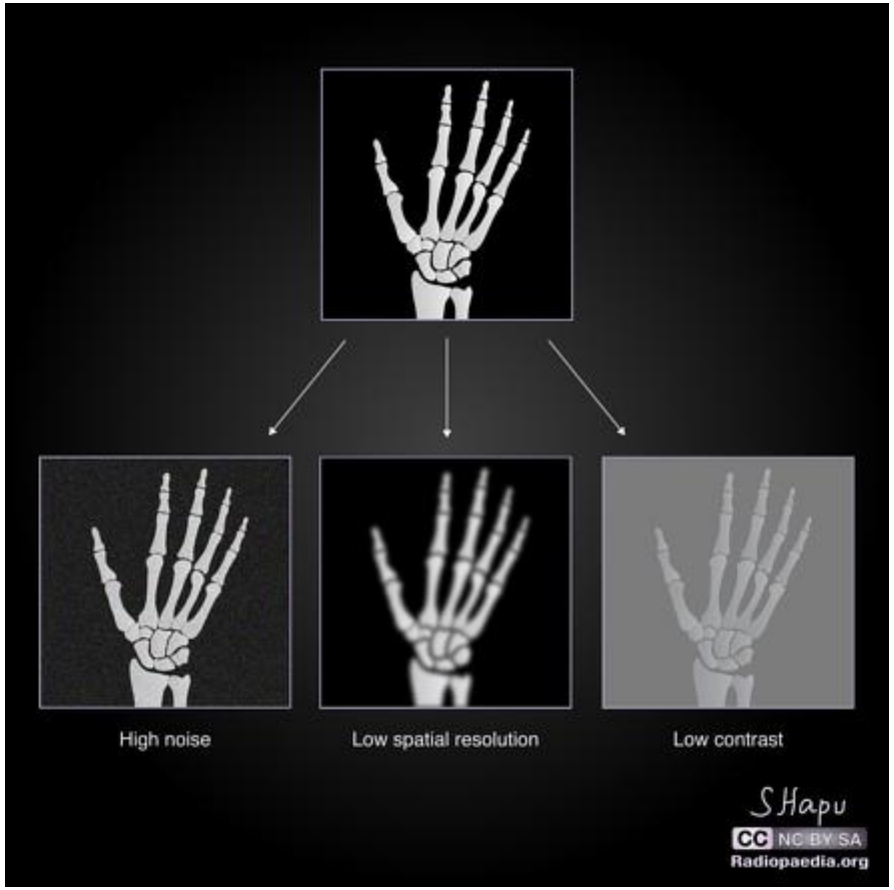
Utility-driven: does the image enable the intended clinical or technical decision?

Key Performance Indicators

Sharpness, Contrast, Noise, and Artifacts — the practical knobs we measure and tune.

Spatial Resolution: The Limit of Detail

✓ Spatial resolution describes the system's ability to distinguish two adjacent high-contrast objects. Clinically, this defines how small a structure can be resolved.



✓ Measurement:

Expressed in line pairs per millimeter (lp/mm).

✓ Influences:

Pixel size, focal-spot size, detector sampling, and slice thickness.

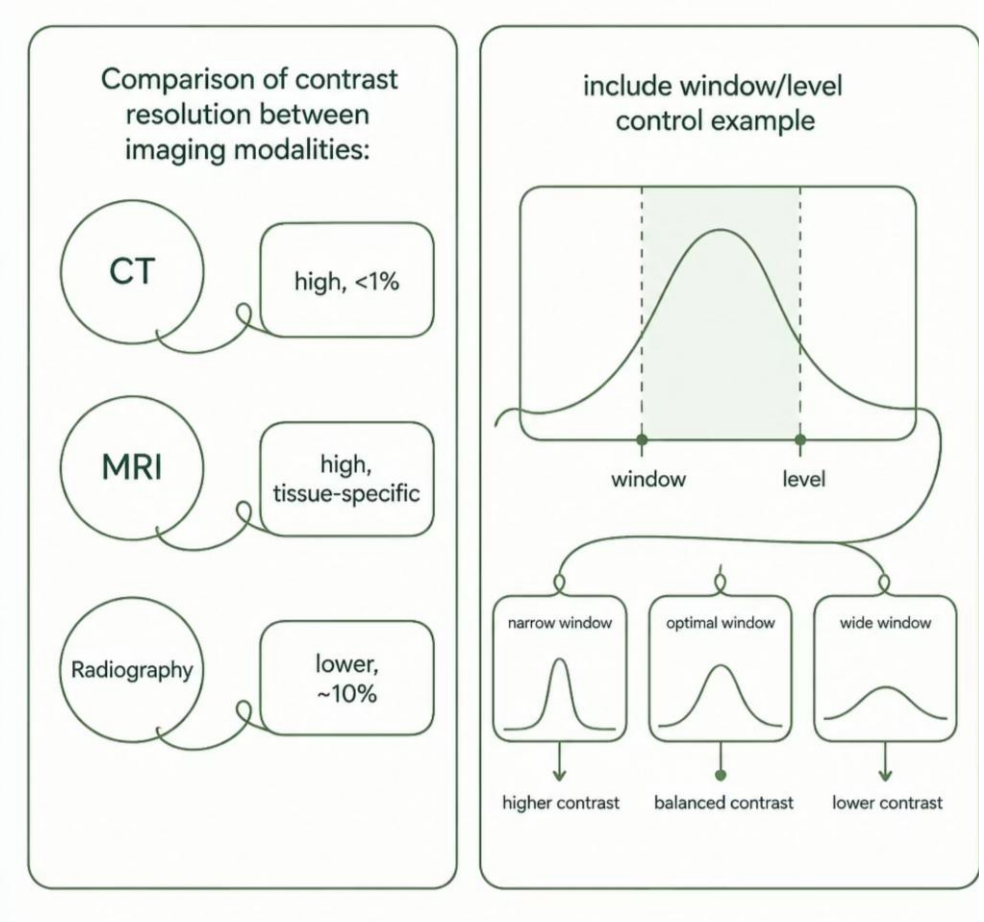
✓ Quantification:

MTF (Modulation Transfer Function) shows contrast preservation across spatial frequencies.

Contrast Resolution: Seeing Beyond Black and White

Contrast resolution is the ability to detect small differences in signal intensity—critical for identifying subtle pathology.

- CT vs radiography: CT routinely detects contrasts <1% while radiography may require ~10% differences.
- Windowing & leveling: Digital tools that remap gray levels to highlight structures of interest.
- Detector dynamic range and A/D precision set the practical contrast limits.



Noise and Artifacts

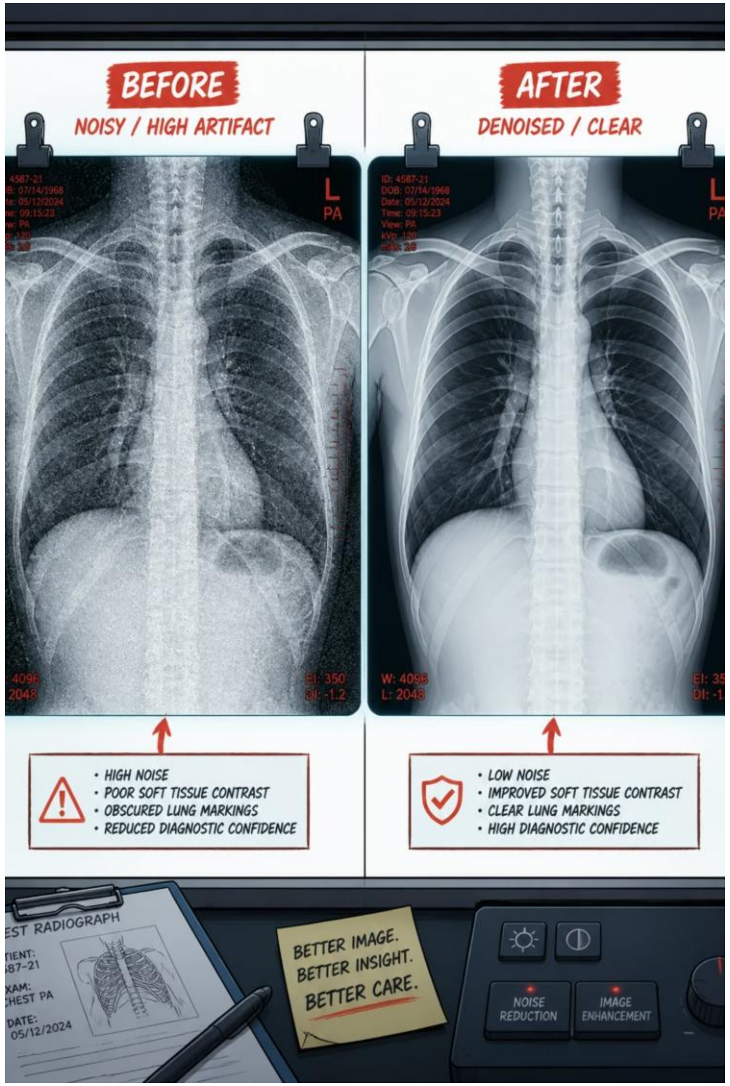
Noise reduces effective contrast and masks low-contrast details. Artifacts introduce misleading structures—both degrade diagnostic confidence.

Noise

SNR (Signal-to-Noise Ratio) is the primary metric; increase mAs or improve detector efficiency to raise SNR.

Common Artifacts

Motion blur, beam hardening (metal streaks), detector saturation (clipping), and blooming from charge leakage.



Geometric & Acquisition Factors

Small changes in geometry and exposure settings produce measurable changes in sharpness and noise.

- Object-to-Detector Distance (OFD): reducing OFD increases sharpness and lowers magnification.
- kVp: higher kVp increases penetration but can reduce subject contrast; adjust based on task.
- mAs: linearly reduces noise when increased, at the cost of dose.
- Collimation: limits scatter, improves contrast and effective sharpness.

Thank You !!